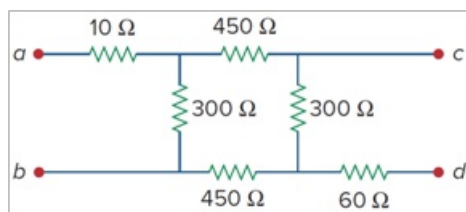


Problem

Consider the circuit in Fig. 2.118. Find the equivalent resistance at terminals: (a) a - b , (b) c - d .

Figure 2.118:



Step-by-step solution

Step 1 of 2

(a)

Consider the circuit of textbook Figure 2.118. For equivalent resistance at terminals a - b , consider terminals c and d open. Therefore don't consider 60Ω resistor. Therefore equivalent resistance is,

$$\begin{aligned} R_{a-b} &= 10 + [300 \parallel (450 + 300 + 450)] \\ &= 10 + [300 \parallel 1200] \\ &= 10 + \left[\frac{(300)(1200)}{1500} \right] \\ &= 10 + 240 \\ &= 250 \Omega \end{aligned}$$

Therefore, the resistance at terminals a - b is $R_{a-b} = 250\Omega$.

Step 2 of 2

(b)

Consider the circuit of textbook Figure 2.118. For equivalent resistance at terminals c - d , consider terminals a and b open. Therefore don't consider 10Ω resistor. Therefore equivalent resistance is,

$$\begin{aligned} R_{c-d} &= 60 + [300 \parallel (450 + 300 + 450)] \\ &= 60 + [300 \parallel 1200] \\ &= 60 + 240 \\ &= 300 \Omega \\ &= 250 \Omega \end{aligned}$$

Therefore, the resistance at terminals c - d is $R_{c-d} = 300\Omega$.